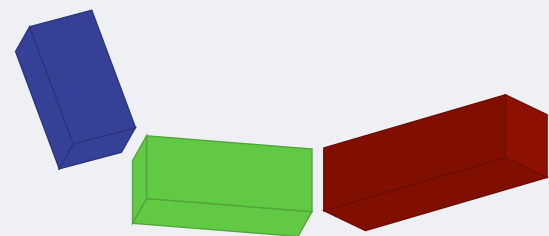


Per-link density — does the 41 g hinge matter?

3-link custom object (v3)

Hinge ignored

estimator has no 41 g hinge body

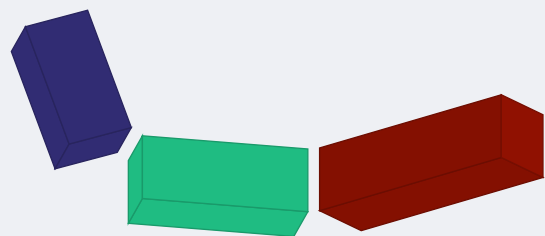


PART	ρ KG/M ³	M KG	ΔM
link0_base	5221.6	1.5044	+0.42%
link1_elbow	1722.2	0.3589	+23.01%
link2_tip	753.1	0.1222	+7.59%
joint1_hinge	—	not modelled	
joint2_hinge	—	not modelled	

claims ±0.38% (2.0 rounds)

Hinge modelled

hinge solved as an extra body

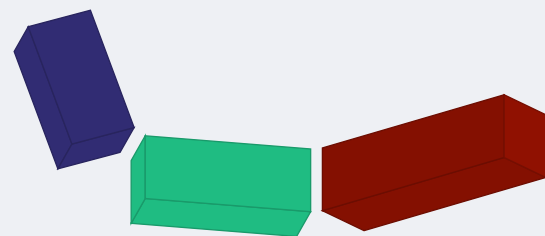


PART	ρ KG/M ³	M KG	ΔM
link0_base	5200.2	1.4983	+0.00%
link1_elbow	1399.8	0.2917	-0.01%
link2_tip	699.9	0.1136	-0.01%
joint1_hinge	1149.2	0.0410	-0.07%
joint2_hinge	1151.0	0.0410	+0.08%

claims ±1.05% (3.2 rounds)

Ground truth

CAD volume × material density



PART	ρ KG/M ³	M KG	ΔM
link0_base	5200.0	1.4982	—
link1_elbow	1400.0	0.2918	—
link2_tip	700.0	0.1136	—
joint1_hinge	1150.0	0.0410	—
joint2_hinge	1150.0	0.0410	—

—

